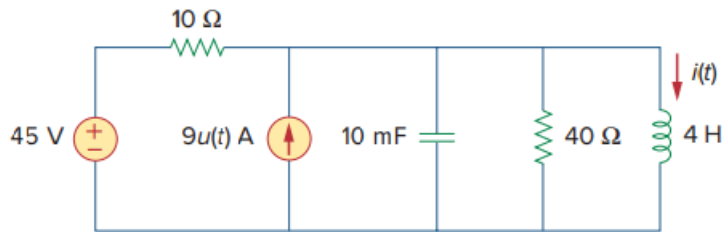


Problem

For the circuit in Fig. 8.97, find $i(t)$ for $t > 0$.

Figure 8.97:



Step-by-step solution

Step 1 of 15

Refer to Figure 8.97 in the text book.

At $t < 0$, the capacitor acts as open circuit and inductor acts as short circuit. The value of the current source is 0 A since the value $u(t)$ at $t < 0$ is 0.

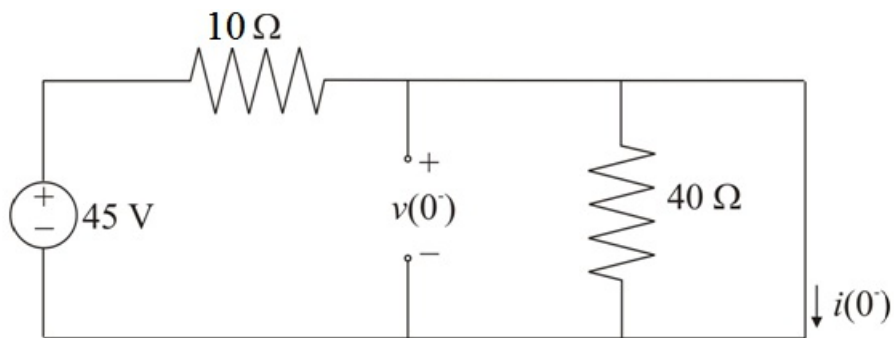


Figure 1

Step 2 of 15

The value of the voltage $v(0^-)$ is 0 V since both the terminals are shorted.

Calculate the value of the current $i(0^-)$.

$$\begin{aligned} i(0^-) &= \frac{45}{10} \\ &= 4.5 \text{ A} \end{aligned}$$

Therefore, the value of the current $i(0^-)$ is 4.5 A.

Step 3 of 15

Draw the circuit diagram at $t > 0$.

The value of the current source is 9 A since $u(t)$ is 1 at $t > 0$.

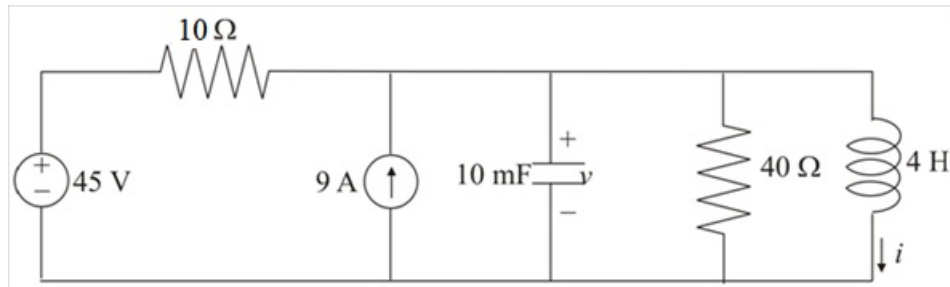


Figure 2

Step 4 of 15

The voltage across the capacitor and the current through the inductor will not change instantaneously. Hence, at $t > 0$ the value of the voltage across the capacitor $v(0^+)$ is 0 V and the value of the current $i(0^+)$ is 4.5 A .

Notice that the voltage across the capacitor and the current through the inductor are continuous. Therefore, the values at $t = 0$ are $v(0) = 0\text{ V}$ and $i(0) = 4.5\text{ A}$.

Step 5 of 15

Calculate the value of the voltage v in terms of the inductance.

$$v = 4 \frac{di}{dt}$$

$$\frac{di}{dt} = \frac{v}{4}$$

Calculate the value of $\frac{di}{dt}$ for $t = 0$.

$$\frac{di(0)}{dt} = \frac{v(0)}{4}$$

Substitute 0 V for $v(0)$.

$$\begin{aligned} \frac{di(0)}{dt} &= \frac{0}{4} \\ &= 0\text{ A/s} \end{aligned}$$

Therefore, the value of $\frac{di(0)}{dt}$ is 0 A/s .

Step 6 of 15

Deactivate the sources from the circuit diagram to estimate the transient response.

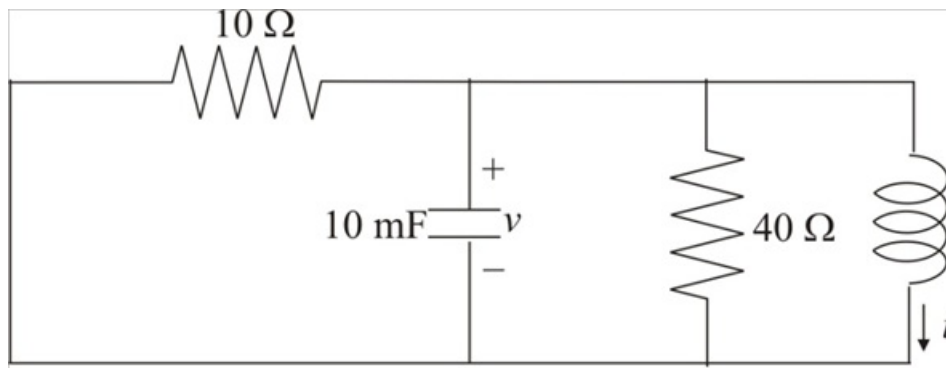


Figure 3

Step 7 of 15

Apply Kirchhoff's Current Law at node-1.

$$\frac{v}{10} + \frac{v}{40} + i + (10 \text{ m}) \frac{dv}{dt} = 0$$

$$\frac{4v + v}{40} + i + (10 \text{ m}) \frac{dv}{dt} = 0$$

$$\frac{5v}{40} + i + (10 \text{ m}) \frac{dv}{dt} = 0$$

$$\frac{v}{8} + i + (10 \text{ m}) \frac{dv}{dt} = 0$$

Substitute $4 \frac{di}{dt}$ for v .

$$\frac{1}{8} \left(4 \frac{di}{dt} \right) + i + (10 \text{ m}) \frac{d}{dt} \left(4 \frac{di}{dt} \right) = 0$$

$$\frac{1}{2} \frac{di}{dt} + i + 0.04 \frac{d^2 i}{dt^2} = 0$$

$$\frac{d^2 i}{dt^2} + 12.5 \frac{di}{dt} + 25i = 0$$

Treat $\frac{d}{dt}$ as s .

$$s^2 i + 12.5s i + 25i = 0$$

$$(s^2 + 12.5s + 25)i = 0$$

Step 8 of 15

Consider the characteristic equation of the circuit.

$$s^2 + 12.5s + 25 = 0$$

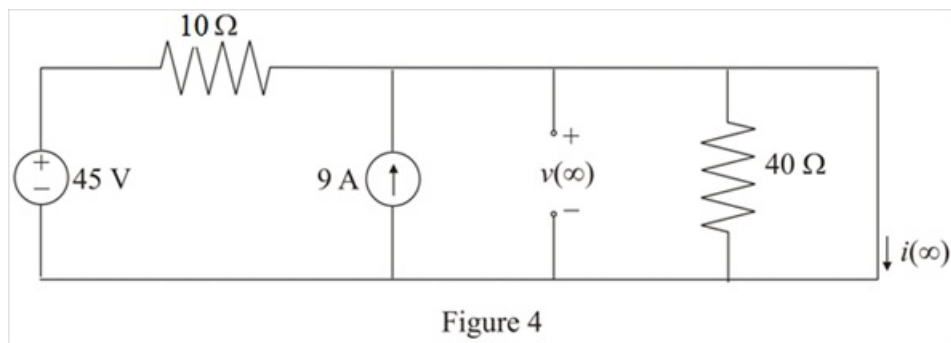
$$(s + 12.5)(s + 25) = 0$$

Therefore, the roots of the equation are -2.5 and -10 .

The roots are negative and real, hence the circuit exhibits over damped response.

Step 9 of 15

Draw the circuit diagram for $t = \infty$.



Step 10 of 15

Consider the value the current $i(\infty)$.

$$\begin{aligned} i(\infty) &= 9 + \frac{45}{10} \\ &= 9 + 4.5 \\ &= 13.5 \text{ A} \end{aligned}$$

Therefore, the value of the current $i(\infty)$ is 13.5 A.

Step 11 of 15

Consider the expression of the current $i(t)$.

$$i(t) = i(\infty) + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

Substitute 13.5 A for $i(\infty)$, -10 for s_1 and -2.5 for s_2 .

$$i(t) = 13.5 + A_1 e^{-10t} + A_2 e^{-2.5t} \dots\dots (1)$$

Step 12 of 15

Recall equation (1).

$$i(t) = 13.5 + A_1 e^{-10t} + A_2 e^{-2.5t}$$

Substitute 0 for t .

$$\begin{aligned} i(0) &= 13.5 + A_1 e^{-10(0)} + A_2 e^{-2.5(0)} \\ &= 13.5 + A_1 e^0 + A_2 e^0 \\ &= 13.5 + A_1 (1) + A_2 (1) \\ &= 13.5 + A_1 + A_2 \end{aligned}$$

Substitute 4.5 A for $i(0)$.

$$4.5 = 13.5 + A_1 + A_2$$

$$A_2 = 4.5 - 13.5 - A_1$$

$$A_2 = -9 - A_1 \dots\dots (2)$$

Step 13 of 15

Recall equation (1).

$$i(t) = 13.5 + A_1 e^{-10t} + A_2 e^{-2.5t}$$

Apply derivative with respect to t .

$$\begin{aligned}\frac{di(t)}{dt} &= A_1 \frac{d(e^{-10t})}{dt} + A_2 \frac{d(e^{-2.5t})}{dt} \\ &= -10A_1 e^{-10t} - 2.5A_2 e^{-2.5t}\end{aligned}$$

Substitute 0 for t and $-9 - A_1$ for A_2 .

$$\begin{aligned}\frac{di(0)}{dt} &= -10A_1 e^{-10(0)} - 2.5(-9 - A_1) e^{-2.5(0)} \\ &= -10A_1 e^0 - 2.5(-9 - A_1) e^0 \\ &= -10A_1(1) - (-22.5 - 2.5A_1)(1) \\ &= -10A_1 + 22.5 + 2.5A_1\end{aligned}$$

Substitute 0 A/s for $\frac{di(0)}{dt}$

$$0 = -7.5A_1 + 22.5$$

$$7.5A_1 = 22.5$$

$$A_1 = \frac{22.5}{7.5}$$

$$A_1 = 3$$

Therefore, the value of the constant A_1 is 3.

Step 14 of 15

Recall equation (2).

$$A_2 = -9 - A_1$$

Substitute 3 for A_1 .

$$\begin{aligned}A_2 &= -9 - 3 \\ &= -12\end{aligned}$$

Therefore, the value of the constant A_2 is -12.

Step 15 of 15

Recall equation (1).

$$i(t) = 13.5 + A_1 e^{-10t} + A_2 e^{-2.5t}$$

Substitute 3 for A_1 and -12 for A_2 .

$$i(t) = 13.5 + 3e^{-10t} - 12e^{-2.5t} \text{ A}$$

Therefore, the value of the current $i(t)$ is $\boxed{13.5 + 3e^{-10t} - 12e^{-2.5t} \text{ A}}$.